Estimating profitability decomposition frameworks via machine learning: Implications for earnings forecasting and financial statement analysis

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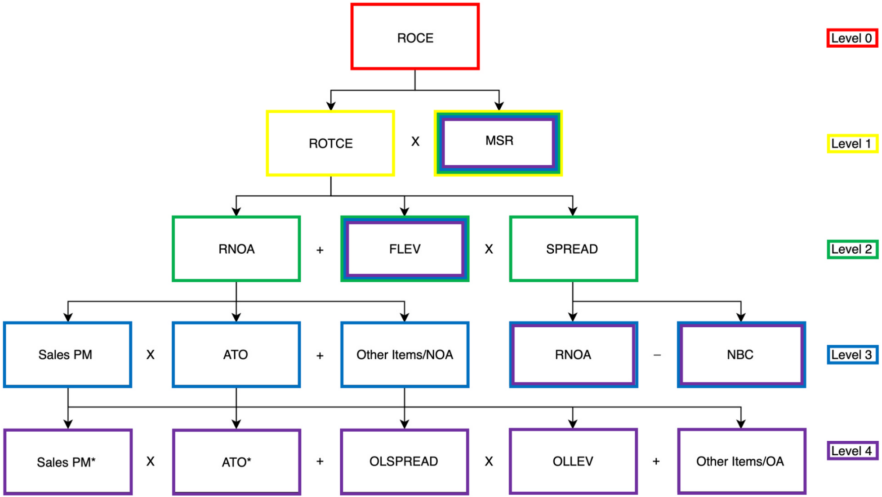
Overview


ROCE
Level 0
ROTCE
X
MSR
Level 1
RNOA
+
FLEV
X
SPREAD
Level 2
Sales PM
X
ATO
+
Other Items/NOA
RNOA
–
NBC
Level 3
Sales PM*
X
ATO*
+
OLSPREAD
X
OLLEV
+
Other Items/OA
Level 4

Motivation

- Hierarchical nonlinear profitability decomposition framework
  - Hierarchical-decomposition to predict profitability foundational in FS analysis.
  - Profitability can be decomposed in many ways: DuPont, ABC, NP
  - NP: structure of accounting equality discipline financial statement information to reveal hierarchy of relation between profitability and accounting drivers.
- Why incorporate profitability structure into machine learning?
  - Adjust for conditions not present in the training dataset (Nissim, 2024)
  - Reduce overlooking value-relevant information in distorted numbers (Sloan, 2019)
  - Protect from fluctuation in numbers induced by reporting process (Penman, 2010)

Research Question

- Whether and how nonlinear machine learning estimation of hierarchical nonlinear profitability decomposition frameworks improves profitability forecast accuracy?
  - Nolinear ML better?
  - Including NP decomposition better?
  - Other ML improve?
  - Other predictor improve?
  - Lag data improve?
  - How compare to investor and analyst?

Contribution

- Contribute to literature analyzing structural profitability decomposition framework.
  - Prior: OLS to approximate nonlinearities in NP's model (Fairfield and Yohn, 2001; Soliman, 2008; Esplin et al.,2014).
  - This paper: ML accommodate nonlinearities improve forecast accuracy.
- Contribute to test whether ML can used to increase earning forecast accuracy.
  - Prior: statistical approach to predictor selection (Anand et al., 2020; Chen et al., 2022; Hunt et al., 2022; Cao and You, 2023; van Binsbergen et al., 2023).
  - Limit: data-mining can produce good predictive performance but difficult to explain.
  - This paper: NP structured decomposition framework guide predictor selection and enhance interpretability of ML.

Nissim and Penman's Profitability Decomposition Framework

- Disaggregate ROCE into four level of cumulative disaggregation
  - **Level 1.** split ROCE into tangible capital return and equity structure adjustment

$$ROCE = ROTCE \times MSR$$

  - **Level 2.** introduce structural information, ROTCE into operate and finance component

$$ROCE = (RONA + FLEV \times SPREAD) \times MSR$$

  - **Level 3.** RNOA into margin and turnover and SPREAD into RNOA and financing cost

$$ROCE = \left[Sales\ PM \times ATO + \frac{Other\ Items}{NOA} + FLEV \times (RNOA - NBC) \right] \times MSR$$

  - **Level 4.** split out operating leverage, capture risk and amplifying effect

$$ROCE = \left[Sales\ PM^* \times ATO^* + \frac{Other\ Items}{OA} + OLLEV \times OLSPREAD \right.$$
$$\left. + FLEV \times (RNOA - NBC) \right] \times MSR$$

Algorithm and Forecasting Procedure

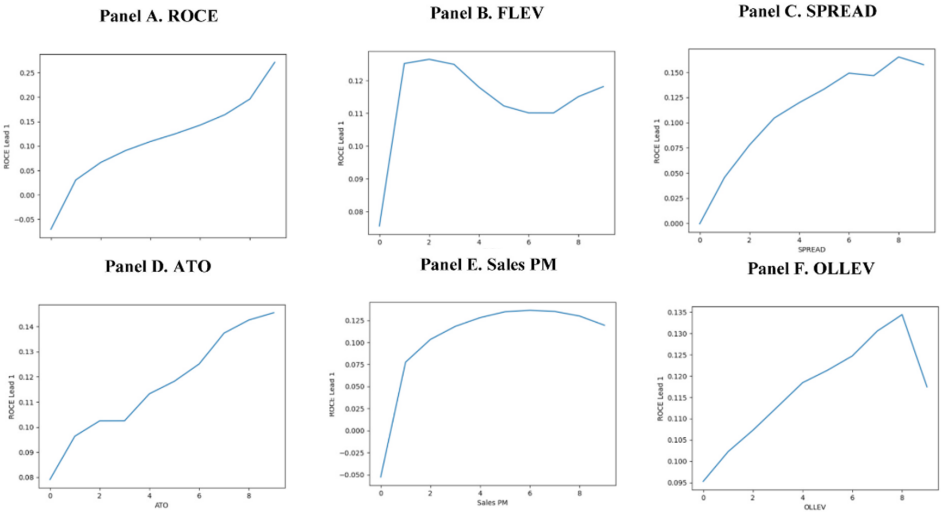
- Variable obtained from four-level decomposition input into $\text{ROCE}_{t+1} = f(X_t)$
  - Level 0: $X_t = (\text{ROCE}_t)$
  - Level 1: $X_t = (\text{ROTCE}_t, \text{MSR}_t)$
  - Level 2: $X_t = (\text{RNOA}_t, \text{FLEV}_t, \text{SPREAD}_t, \text{MSR}_t)$
  - Level 3: $X_t = (\text{Sales PM}_t, \text{ATO}_t, \text{Other Items NOA}_t, \text{FLEV}_t, \text{RNOA}_t, \text{NBC}_t, \text{MSR}_t)$
  - Level 4: $X_t = (\text{Sales PM}*_t, \text{ATO}*_t, \text{Other Items OA}_t, \text{OLLEV}_t, \text{OLSPREAD}_t, \text{FLEV}_t ...)$
- $f(\cdot)$ include 5 ML models: random walk, linear OLS, nonlinear GBRT, NN and RF
  - Random walk yield more accurate prediction than linear estimation of complex earning forecast model (Gerakos and Gramacy, 2013; Li and Mohanram, 2014)

Data and Sample Selection

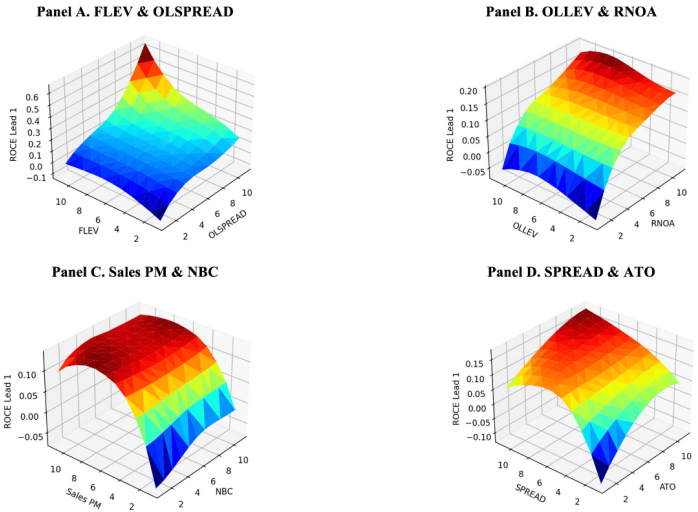
- Financial data from Compustat; Stock market data from CRSP (1963 - 2023)
- Sample selection to ensure data quality and empirical validity
  - Ratio in NP framework for current year and ROCE for next year complete
  - Liquidity and size requirement (market capitalization higher than 20th percentile)
  - Meet accounting constraint (CSE, NOA, OA, MIB, OL non-negative)

	N	Mean	SD	P1	P25	Median	P50	P99
ROCE	117,186	0.10	0.21	−0.70	0.06	0.12	0.18	0.69
ROTCE	117,186	0.11	0.22	−0.70	0.06	0.12	0.18	0.70
MSR	117,186	0.99	0.05	0.75	1.00	1.00	1.00	1.08
RNOA	117,186	0.14	0.48	−1.19	0.06	0.10	0.18	1.87
FLEV	117,186	0.61	1.46	−0.96	−0.12	0.31	0.89	7.61
SPREAD	117,186	0.10	0.59	−1.66	0.00	0.05	0.16	2.48
Sales PM	117,186	0.06	0.44	−1.10	0.04	0.07	0.14	0.61
ATO	117,186	2.19	2.57	0.12	0.79	1.56	2.55	14.08
Other Items NOA	117,186	0.00	0.01	−0.02	0.00	0.00	0.00	0.07
NBC	117,186	0.04	0.13	−0.48	0.01	0.04	0.06	0.55
*Sales PM**	117,186	0.07	0.44	−1.08	0.04	0.08	0.15	0.67
*ATO**	117,186	1.22	0.92	0.06	0.55	1.06	1.61	4.61
OLSPREAD	117,186	0.05	0.18	−0.58	0.01	0.04	0.09	0.63
OLLEV	117,186	0.88	2.33	0.05	0.26	0.40	0.66	12.30
Other Items OA	117,186	0.00	0.01	−0.01	0.00	0.00	0.00	0.04

Nonlinearity between Future Profitability and Contemporaneous Ratio



Interactive Relation across Ratio to Future Profitability



Whether OLS and GBRT Yield More Accurate Prediction?

- Absolute forecast error from estimation of year-ahead profitability in autoregression.
  - Group in deciles based on forecast error of RW and compute each decile's mean error.
  - Compare performance in different prediction difficulty ranges:
  - OLS avoid large forecast errors than frequently make precise forecast.
  - GBRT adapt to different data structure, thus stable across entire distribution.

Error Decile	RW	OLS	GBRT
All	**12.297**	**12.203**	**11.439*** **
1	0.294	0.514	0.302
2	0.930	1.521	0.946
3	1.696	2.561	1.692
4	2.638	3.675	2.584***
5	3.867	4.933	3.716***
6	5.533	6.476	5.230***
7	8.042	8.516	7.422***
8	12.055	11.799***	10.982***
9	20.318	18.884***	18.280***
10	67.593	63.154***	63.230***

Whether Including Four-Level NP Decomposition Improve Accuracy?

Panel A. Linear Estimation (OLS)

Error Decile	Level 0	Level 1	Level 2	Level 3	Level 4
All	**12.203**	**12.198**	**14.279**	**14.101**	**13.648**
1	0.514	0.513	0.637	0.628	0.603
2	1.521	1.516	1.916	1.892	1.813
3	2.561	2.553**	3.252	3.220	3.064
4	3.675	3.662***	4.691	4.647	4.411
5	4.933	4.910***	6.345	6.265	5.960
6	6.476	6.454***	8.426	8.256	7.848
7	8.516	8.501*	11.118	10.880	10.348
8	11.799	11.804	15.076	14.736	14.157
9	18.884	18.894	22.635	22.089	21.542
10	63.154	63.173	68.690	68.397	66.730

Panel B. Nonlinear estimation (GBRT)

Error decile	Level 0	Level 1	Level 2	Level 3	Level 4
All	**11.439**	**11.437**	**11.314**	**11.192****	**11.051*****
1	0.302	0.307	0.311	0.312	0.308
2	0.946	0.948	0.954	0.970	0.958
3	1.692	1.693	1.694	1.713	1.695
4	2.584	2.585	2.578*	2.596	2.583
5	3.716	3.717	3.698***	3.699***	3.675***
6	5.230	5.230	5.202***	5.195***	5.141***
7	7.422	7.421	7.365***	7.317***	7.241***
8	10.982	10.973	10.857***	10.719***	10.604***
9	18.280	18.275	17.957***	17.666***	17.402***
10	63.230	63.214	62.521	61.733**	60.903***

Whether Nonlinear Structure Generally Effective?

Panel A. Comparison to RW

Error Decile	RW	Neural Network	Random Forest	Holdout
All	**12.297**	**11.437*****	**11.428*****	**11.457*****
1	0.294	0.306	0.301	0.303
2	0.930	0.954	0.940	0.942
3	1.696	1.695	1.683***	1.686***
4	2.638	2.585***	2.576***	2.579***
5	3.867	3.710***	3.703***	3.708***
6	5.533	5.211***	5.220***	5.233***
7	8.042	7.393***	7.412***	7.432***
8	12.055	10.921***	10.983***	10.996***
9	20.318	18.195***	18.298***	18.352***
10	67.593	63.393***	63.157***	63.332***

Panel B. NP Variables

	Neural Network		Random Forest		Holdout	
Error Decile	Level 0	Level 4	Level 0	Level 4	Level 0	Level 4
All	**11.437**	**11.187****	**11.428**	**11.143*****	**11.457**	**11.168*****
1	0.306	0.328	0.301	0.304	0.303	0.310
2	0.954	1.018	0.940	0.946	0.942	0.971
3	1.695	1.793	1.683	1.692	1.686	1.715
4	2.585	2.692	2.576	2.573	2.579	2.607
5	3.710	3.804	3.703	3.692**	3.708	3.714
6	5.211	5.264	5.220	5.185***	5.233	5.182***
7	7.393	7.369**	7.412	7.317***	7.432	7.272***
8	10.921	10.739***	10.983	10.738***	10.996	10.638***
9	18.195	17.627***	18.298	17.632***	18.352	17.541***
10	63.393	61.237***	63.157	61.349**	63.332	61.725**

Can Adding Other Predictor Further Improve Accuracy?

- Compare performance of NP to 5 types of extra variables:
  - (1) Dupont analysis
  - (2) Hou, Van Dijk, and Zhang's (2012) profitability forecasting model
  - (3) Fama and French (1997) 48 industry fixed effect to NP
  - (4) Real GDP growth, inflation, and unemployment rate to NP
  - (5) industry fixed effect and macroeconomic variable to NP

Error Decile	Baseline	Dupont	Hou	Industry	Macro	Industry & Macro
All	**11.376**	**11.722**	**12.340**	**11.464**	**11.405**	**11.388**
1	0.317	0.350	0.416	0.329	0.330	0.336
2	0.980	1.076	1.276	1.016	1.014	1.031
3	1.742	1.893	2.188	1.792	1.783	1.801
4	2.645	2.851	3.229	2.706	2.696	2.716
5	3.768	4.042	4.477	3.857	3.834	3.855
6	5.275	5.617	6.091	5.375	5.339	5.366
7	7.432	7.826	8.337	7.548	7.483	7.505
8	10.892	11.333	11.899	11.047	10.929	10.947
9	17.907	18.440	19.336	18.140	17.918	17.898
10	62.803	63.792	66.147	62.828	62.724	62.426

Does Retaining Only Core Items Improve Accuracy?

- Trade-off between information loss and prediction-irrelevant noise
  - Core items persistent, directly related to profitability
  - Non-core items transitory (mean reverting), deriving from unusual operating income
  - Current-period impairment loss increase future benefit (Penman and Zhang, 2002)
  - Manager manipulate income presentation to blur core/non-core distinct (McVay, 2006)

Error Decile	Baseline	Core
All	**11.051**	**10.677*** **
1	0.308	0.309
2	0.958	0.949***
3	1.695	1.650***
4	2.583	2.484***
5	3.675	3.523***
6	5.141	4.886***
7	7.241	6.833***
8	10.604	10.037***
9	17.402	16.636***
10	60.903	59.464**

How Much Historical Information to Consider?

- Trade-off between more information and stale information
  - More lags capture how profitability components behave through business cycle.
  - Longer period increase likelihood firm's activities have changed sufficiently.

Error Decile	Baseline	Lag 1	Lag 2	Lag 3
All	**9.732**	**9.691**	**9.665**	**9.673**
1	0.294	0.289**	0.284***	0.290*
2	0.905	0.903	0.896***	0.901*
3	1.609	1.595***	1.592***	1.599***
4	2.442	2.433**	2.434**	2.433**
5	3.490	3.473***	3.471***	3.473***
6	4.846	4.836	4.803***	4.813***
7	6.804	6.722***	6.699***	6.704***
8	9.812	9.658***	9.616***	9.613***
9	15.675	15.446***	15.369***	15.350***
10	51.439	51.552	51.482	51.558

Can Forecast from NP Inform Investor and Analyst?

Panel A. Stock Returns

	Returns					
	(1)	(2)	(3)	(4)	(5)	(6)
Optimal Model	0.255***	0.247***	0.256***	0.248***	0.258***	0.251***
	(3.028)	(2.868)	(3.028)	(2.870)	(3.074)	(2.915)
Median Analyst Forecast			−0.001	−0.001		
			(-0.422)	(-0.476)		
Mean Analyst Forecast					−0.005	−0.005
					(-1.322)	(-1.335)
Beta		−0.003		−0.003		−0.003
		(-0.164)		(-0.165)		(-0.166)
SMB		0.009		0.009		0.009
		(0.437)		(0.435)		(0.430)
HML		0.022**		0.022**		0.022**
		(2.177)		(2.177)		(2.178)
RMW		−0.022		−0.022		−0.022
		(-1.270)		(-1.270)		(-1.270)
CMA		0.013		0.013		0.013
		(1.109)		(1.110)		(1.112)
UMD		−0.017		−0.017		−0.017
		(-0.581)		(-0.580)		(-0.580)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year-Month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	19,050	19,050	19,050	19,050	19,050	19,050
R-squared	0.464	0.466	0.464	0.466	0.464	0.466

Can Forecast from NP Provide Incremental Information?

Panel B. Changes in profitability

	ΔROCE					
	(1)	(2)	(3)	(4)	(5)	(6)
Optimal Model	1.058***	1.058***	1.051***	1.050***	1.046***	1.045***
	(23.638)	(23.714)	(22.536)	(22.600)	(23.663)	(23.716)
Median Analyst Forecast			0.010	0.010		
			(1.395)	(1.405)		
Mean Analyst Forecast					0.017*	0.017*
					(1.974)	(1.977)
Beta		0.003		0.003		0.003
		(0.353)		(0.369)		(0.368)
SMB		−0.001		−0.001		−0.001
		(-0.276)		(-0.199)		(-0.158)
HML		−0.002		−0.002		−0.002
		(-0.387)		(-0.414)		(-0.438)
RMW		0.005		0.005		0.005
		(1.105)		(1.104)		(1.103)
CMA		0.000		0.000		−0.000
		(0.029)		(0.004)		(-0.016)
UMD		−0.004		−0.004		−0.005
		(-0.615)		(-0.618)		(-0.622)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year-Month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	19,050	19,050	19,050	19,050	19,050	19,050
R-squared	0.443	0.443	0.444	0.444	0.445	0.445

Discussion

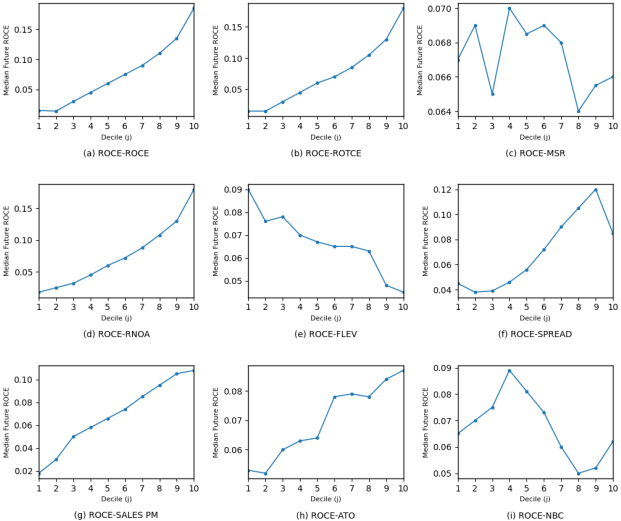
- Theory-guided / structured ML
- (1) Theory determine input structure
  - Instead of massive variables, only using theoretical variables to control variable structure.
- (2) Theory introduce model constraint
  - Modeling layer by layer according to theory
  - Multi-task learning to simultaneously predict intermediate variables in thoery
  - Incorporating economic constraints (monotonicity, structure regularization)
- (3) Theory as interpretation
  - SHAP
  - Partial dependence

中国市场的机器学习估算盈利能力分解框架

- 以 2006 至 2025 年我国 A 股上市公司为样本，数据来源于 CSMAR。
  - 为保证数据质量与实证有效性，样本需满足：（1）当年 NP 框架全部变量不缺失；（2）下一年 ROCE 不缺失；（3）市值位于 A 股市场前一半；（4）具有证监会行业代码；（5）总资产大于 1 亿人民币；（6）会计约束条件（CSE、NOA、OA、MIB、OL 非负）。
  - 最终得到 21498 个企业-年度观测值。

	count	mean	std	min	25%	50%	75%	max
ROCE	21498	0.04	0.33	-46.52	0.02	0.06	0.12	10.55
ROTCE	21498	0.03	0.38	-25.75	0.01	0.05	0.09	20.20
MSR	21498	0.08	0.16	0.00	0.00	0.03	0.10	10.72
RNOA	21498	0.05	0.28	-25.46	0.03	0.07	0.12	13.88
FLEV	21498	0.01	1.21	-24.21	-0.29	-0.08	0.12	77.14
SPREAD	21498	0.07	5.95	-139.15	-0.07	0.04	0.15	687.30
SALES PM	21498	0.26	5.64	-345.42	0.03	0.08	0.15	673.78
ATO	21498	1.22	2.06	0.00	0.42	0.77	1.19	128.37
Other Items/NOA	21498	0.03	0.32	-7.28	0.00	0.01	0.03	19.28
NBC	21498	-0.02	6.10	-619.62	-0.03	0.01	0.10	144.55

未来盈利能力与同期比率之间的非线性关系和交互关系



(a) ROCE-ROCE (b) ROCE-ROTCE (c) ROCE-MSR

(d) ROCE-RNOA (e) ROCE-FLEV (f) ROCE-SPREAD

(g) ROCE-SALES PM (h) ROCE-ATO (i) ROCE-NBC

自回归估计预测误差对比

Decile	RW	OLS	GBRT
1	0.162	0.365	0.546
2	0.534	1.081	0.924
3	1.028	1.919	1.232
4	1.577	2.821	1.448
5	2.297	3.510	2.176
6	3.310	5.106	2.807***
7	4.670	7.079	3.725***
8	6.943	10.163	5.551***
9	12.117	15.094	9.737***
10	53.819	48.925**	40.820***
ALL	8.123	9.391	6.652***

NP 框架估计预测误差对比

Panel A. OLS

Decile	Level 0	Level 1	Level 2	Level 3
1	0.365	0.700	0.583	0.727
2	1.081	1.275	1.061	1.254
3	1.919	2.185	1.914	2.209
4	2.821	2.855	2.701	2.692
5	3.510	3.789	3.635	3.691
6	5.106	4.974	5.063	4.963
7	7.079	7.560	7.003	7.200
8	10.163	9.947	10.310	9.854
9	15.094	16.979	16.912	16.504
10	48.925	47.950	50.901	49.084
ALL	9.391	9.995	9.430	10.243

Panel B. GBRT

Decile	Level 0	Level 1	Level 2	Level 3
1	0.546	0.569	1.580	1.944
2	0.924	0.891	1.152	1.325
3	1.232	1.227	1.000***	1.240
4	1.448	1.464	1.012***	1.081***
5	2.176	2.150	1.931***	1.715***
6	2.807	2.882	2.188***	2.236***
7	3.725	4.079	2.691***	2.289***
8	5.551	6.024	3.623***	3.058***
9	9.737	9.076	6.152***	5.183***
10	40.820	39.876	35.067***	33.434***
ALL	6.652	6.642	5.831***	5.586***

不同机器学习方法误差对比

Decile	RW	NN	RF	GBRT
1	0.162	0.168	0.171	0.546
2	0.534	0.541	0.549	0.924
3	1.028	1.012	1.020	1.232
4	1.577	1.555	1.562	1.448
5	2.297	3.170	2.285	2.176
6	3.310	2.680	2.730**	2.807***
7	4.670	3.420**	3.510**	3.725***
8	6.943	4.580***	4.690***	5.551***
9	12.117	7.360***	7.520***	9.737***
10	53.819	36.850***	37.480***	40.820***
ALL	8.123	6.710***	6.800***	6.652***

引入滞后项

Decile	Baseline	Lag 1	Lag 2	Lag 3
1	0.546	0.541*	0.537**	0.542*
2	0.924	0.918	0.912**	0.919*
3	1.232	1.221**	1.218***	1.224***
4	1.448	1.436**	1.433**	1.435**
5	2.176	2.158**	2.151**	2.156**
6	2.807	2.786	2.765**	2.772***
7	3.725	3.680**	3.655**	3.668***
8	5.551	5.470***	5.420***	5.435***
9	9.737	9.600***	9.540***	9.565***
10	40.820	40.950	40.880	41.020
ALL	6.652	6.610**	6.585**	6.598***